Mahmud Hasin

Dhaka, Bangladesh | (+880) 1973-551947 | mahmudhasin4@gmail.com

OBJECTIVE

Highly motivated Mechanical Engineering undergraduate specializing in molecular dynamics, computational modeling, and machine-learning-accelerated simulation. Experienced in modeling atomic-scale phenomena and finite-element convergence behavior using LAMMPS, COMSOL, and deep learning frameworks. Currently investigating ML-accelerated multiscale modeling for HVAC thermal systems for my undergraduate thesis. Seeking opportunities to advance research at the intersection of computational materials science, deep learning, and simulation-driven engineering design.

EDUCATION

Ahsanullah University of Science and Technology (AUST) April 2023 – May 2027 (Expected)
B.Sc., Mechanical Engineering (ME) *Dhaka, Bangladesh*

CGPA: 3.09/4.0 (Current); Projected: ~3.15/4.0 after graduation

Relevant Coursework: Machine Learning, Deep Learning, Computational Mechanics, Numerical Methods, CAD & Simulation, Engineering Mathematics, MS Office Suite

RESEARCH INTERESTS

Molecular Dynamics · Machine-Learning-Accelerated Modeling · Deep-Generative & Energy-Based Methods for Materials Design · Finite Element Analysis · Computational Fluid Dynamics

RESEARCH EXPERIENCE

ML-Accelerated Multiscale Modeling for Designing an Optimal HVAC Cooling Element 2026 – Present
Dept. of Mechanical Engineering, AUST, Dhaka, Bangladesh

- Supervisor: Prof. Dr. Md. Kharshiduzzaman
- Developing a machine-learning-accelerated multiscale simulation framework to optimize thermal performance of an HVAC cooling element; undergraduate thesis, in progress.

Deep Learning-Based Finite Element Convergence Analysis Ongoing
AUST Applied Simulation and Modeling Lab (AASML), Dhaka, Bangladesh

- Developed convolutional neural networks (CNNs) to analyze output images from finite element method (FEM) simulations and evaluate numerical accuracy across progressively refined meshes.
- Integrated Lagrange elements in COMSOL to automate convergence analysis, improving efficiency in numerical simulations.

Molecular Dynamics Simulation of Droplet Impact and Evaporation (LAMMPS) Ongoing
AUST Applied Simulation and Modeling Lab (AASML), Dhaka, Bangladesh

- Simulated a liquid droplet impacting a solid surface, modeling droplet–droplet and droplet–wall interactions using Lennard-Jones potentials.
- Designed a 3D molecular dynamics system in a 40×40×40 simulation box and simulated droplet evaporation under controlled thermal loading (0.5 → 2.0 LJ units), tracking coordination number and kinetic energy.

PUBLICATIONS

Conference Proceedings

- Mahmud Hasin. A Solar-Powered Perpetual Mars Exploration Unmanned Aerial Vehicle: A Review of Sky-Sailor. Accepted for presentation, Symposium on Photonics, Emerging Computational Technologies, Research & AI-Data Science (SPECTRA 2026), Open Category. 2026.
- Mahmud Hasin. A Study on Molecular Dynamics for the Deep Generative Inverse Design of Gold Nanowires. Accepted for oral presentation, 8th International Conference on Chemical Engineering (ICChE 2026), Paper ID 171. 2026.

Under Review / In Preparation

- Coarse-Grained Boltzmann Generators for Data-Efficient Molecular Sampling Using Energy-Based Training. ICMIME 2026 (in preparation).
- Machine Learning Integrated with Molecular Dynamics for Predicting Interfacial Interaction Energies of Ciprofloxacin–Carbon Nanotube Systems. Material Science Track, Submission ID 223 (awaiting decision).
- Machine Learning Integrated Molecular Dynamics for Predicting Orientation-Dependent Deformation Behavior of Single-Crystalline Gold Nanowires. Material Science Track, Submission ID 239 (awaiting decision).
- Orientation-Dependent Tensile Properties and Atomic-Scale Deformation Mechanisms of Single-Crystal Fe, Ni, and Fe–Ni Alloys: A Molecular Dynamics Study. Material Science Track, Submission ID 248 (awaiting decision).

APPLICABLE SKILLS

Programming Python (TensorFlow, PyTorch, Keras, NumPy, Pandas, Scikit-learn), Django, MATLAB, C

Simulation & Modeling LAMMPS, OpenMM, COMSOL, Finite Element Analysis (FEM), CAD (SolidWorks)

Data Analysis & Viz MATLAB, Matplotlib, Seaborn, OVITO

Documentation LaTeX (technical reports, conference papers, presentations), Microsoft Office

LEADERSHIP AND SERVICE

Machine Learning Intern, FlyRank AI July 2026 – Present

Dhaka, Bangladesh

- Contributing to applied machine learning initiatives as part of the AI team.

Intern, Dividend Management — DividendU Program May 2026 – August 2026

- Selected for DividendU, Dividend Management's structured summer internship program.

Django Developer, Perpex April 2026 – Present

- Developing and maintaining backend features using the Django web framework.

Artificial Intelligence Intern, Zetheta Algorithms Private Limited March 2026 – Present

- Supporting AI-focused research and development initiatives within the organization.

Sub-Executive, Content Writing, AUST Research and Publication Club March 2024 – September 2024

- Produced content supporting the club's research-dissemination and publication initiatives.

INDEPENDENT PROJECTS

Matsonnet Studio — AI Workspace for Molecular Dynamics

- Built an AI-powered research workspace centered on OpenMM for molecular dynamics simulation, allowing users to describe a molecular system in plain language and receive a fully configured simulation, trajectory, and analysis in return.
- Designed an agentic, natural-language interface to lower the barrier to entry for running and interpreting molecular dynamics simulations.

CERTIFICATIONS AND COURSES

- Certified Peer Reviewer Course.
- Deep Learning Specialization, Andrew Ng — Coursera / DeepLearning.AI — neural networks, CNNs, and sequence model fundamentals.

REFERENCES

Available upon request.